

E. Beautiful Subarrays

time limit per test: 3 seconds

memory limit per test: 512 megabytes

One day, ZS the Coder wrote down an array of integers a with elements $a_1, a_2, \dots, a_n$.

A subarray of the array a is a sequence $a_l, a_{l+1}, \dots, a_r$ for some integers (l, r) such that $1 \leq l \leq r \leq n$. ZS the Coder thinks that a subarray of a is beautiful if the bitwise xor of all the elements in the subarray is at least k .

Help ZS the Coder find the number of beautiful subarrays of a !

Input

The first line contains two integers n and k ($1 \leq n \leq 10^6, 1 \leq k \leq 10^9$) — the number of elements in the array a and the value of the parameter k .

The second line contains n integers a_i ($0 \leq a_i \leq 10^9$) — the elements of the array a .

Output

Print the only integer c — the number of beautiful subarrays of the array a .

Examples

input	Copy
3 1 1 2 3	
output	Copy
5	
input	Copy
3 2 1 2 3	
output	Copy
3	
input	Copy
3 3 1 2 3	
output	Copy
2	

Educational Codeforces Round 12

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Problem tags

data structures

divide and conquer

strings

trees

*2100

No tag edit access

→ Contest materials

- Announcement (en)

✕
- Editorial (en)

✕

